

Quick Reference — Containers for HPC

GET & CONVERT AN IMAGE

```
apptainer pull docker://org/img          fetch → local .sif
apptainer pull oras://ghcr.io/org/img:tag OCI registry (signed; rate-limit relief)
apptainer build img.sif docker://org/img convert (can add a thin def on top)
apptainer build img.sif my.def           build from a definition file
```

RUN IT

```
apptainer run img.sif                runs the %runscript (default action)
apptainer exec img.sif CMD           run an arbitrary command
apptainer shell img.sif              interactive shell inside
./img.sif                            SIF is executable → same as run
```

BIND DATA & ISOLATE

```
--bind /scratch/$USER              mount a host path
--bind /proj/data:/data:ro         mount read-only (shared datasets)
APPTAINER_BIND=...                 set binds via environment
--no-home / -c / -C                drop home / contain / containall
```

Bound by default: \$HOME, /tmp, /proc, /sys, /dev, \$PWD. **Not** /scratch or project FS — bind them (or set bind path in apptainer.conf).

GPUS

```
apptainer exec --nv img.sif ...      NVIDIA: binds host driver libs
apptainer exec --rocm img.sif ...    AMD ROCm
apptainer exec --nv img.sif nvidia-smi smoke test
```

Host driver bound in; CUDA toolkit lives in the image → portable. Image CUDA must be ≤ host driver's max.

ENVIRONMENT / INSPECT

```
--cleanenv (-e)                    deterministic env (stop host leak)
--env VAR=val                       set a variable
APPTAINERENV_VAR=val               pass a host var through
apptainer inspect [--deffile] img.sif labels / recover the recipe
```

INSTANCES (SERVICES)

```
apptainer instance start img.sif web start named (runs %startscript)
apptainer instance list              what's running
apptainer exec instance://web CMD    act on a running instance
apptainer instance stop web          stop it
```

BUILD & ADVANCED

```
apptainer build --fakeroot img.sif def      unprivileged build (needs subuid/subgid)
apptainer build --build-arg version=9 ...   override %arguments {{ version }}
apptainer build --sandbox dev/ docker://... writable dir for debugging
apptainer build img.sif dev/               freeze a sandbox back to SIF
apptainer overlay create --size 1024 o.img  persistent writable layer over RO SIF
spack containerize > rocky.def             generate a def from a Spack env
```

SIGN & VERIFY (PROVENANCE)

```
apptainer key newpair               generate a PGP keypair
apptainer sign img.sif              sign with your private key
apptainer verify img.sif            verify signature(s)
apptainer key push <fingerprint>   publish public key to a keyserver
```

DEF FILE SKELETON

```
Bootstrap: docker
From: rockylinux:9
%arguments
    version="9"
%files
    src    /opt/dst
%post
    dnf install -y ...
    dnf clean all    # keep image small
%environment        # RUNTIME env vars
    export PATH=/opt/app/bin:$PATH
%runscript
    exec myapp "$@"
%test
    myapp --version
%labels
    Maintainer you@site.edu
```

RUN UNDER SLURM

```
#SBATCH -N 2 --gres=gpu:1
module load apptainer
srun apptainer exec --nv \
    --bind /scratch/$USER \
    /shared/img.sif my_app ...
```

Container inherits the job's cgroup, accounting & limits. Stage big SIFs on fast/local FS.

MPI: THE HANDSHAKE

```
# build MPI with PMI-2 support:
./configure --with-slurm \
    --with-pmi=/usr
# launch (host half must match):
srun --mpi=pmi2 apptainer exec \
    img.sif mpirun ...
```

All ranks = rank 0? PMI handshake failed (missing --mpi=, version skew). **Over TCP?** bind /dev/infiniband + host fabric libs.

ADMIN KNOBS

```
apptainer.conf    allow setuid, bind path, enable fakeroot, allow container
/etc/subuid,subgid  UID/GID ranges → who may --fakeroot build
config fakeroot --add U grant a user fakeroot
```

Decide: build on-cluster (grant subuid) or import-only (signed SIFs). Setuid vs rootless (users) is the core posture choice.

TROUBLESHOOTING

```
command not found      container's PATH → fix %environment
permission denied (write) RO image → --bind a writable dir
GPU not found          forgot --nv / --rocm
wrong/missing library  host lib leaked → --cleanenv
cannot create namespace users disabled → enable or use setuid
```